Equations and inequalities

Teacher resource

Question 1

$a^2 = b$

Which of the following could **not** be true?

O a = -3 and b = 9

O a = 1 and b = 1

O a = -1 and b = 1

O a = 9 and b = 3

Answer: D
Skill: Students evaluate expressions for equivalence

Additional questions

1. If a = 4, can $\sqrt{a} = -a \div 2$ be true?
2. If b = 12, can $b + b = b^2 \div 6$ be true?
3. If c = ½, can $c \div 2 = c^2$ be true?

Question 2

$$100 = 26 + \frac{37}{n}$$

What value of n makes this equation true?

O 74

O 64

O 2

O $\frac{1}{2}$

Answer: D
Skill: Students evaluate equations including fractions

Additional questions

1. If a = 2, what is the value of $\frac{2a}{2a+1}$?
2. If x = 2, what is the value of $\frac{2x}{1+2x}$?
3. What is the even number closest to the value of $\frac{2a}{2a+1}$, when a = 0.5?

Question 3

$3x^2 < 81$

How many possible values can x have if x is a positive integer?

O 3

O 5

O 6

O 9

Answer: 5
Skill: Students evaluate inequalities.

Additional questions

1. How many possible values can x have if x is a negative integer?
2. How many possible values can x have if x is an integer and $3x^2 = 81$?
3. How many possible values can x have if $1 < x < 10$, x is an integer and $3x^2 > 81$?

Question 4

$$-200 > \frac{5a}{4}$$

Which statement must be true?

O a = -160

O a > -160

O a < -160

O a = 160

Answer: C
Skill: Students evaluate inequalities

Additional questions

1. Name a pair of values a and b, where a and b are negative integers, that make $\frac{1}{a} > \frac{1}{b}$ true.

2. Name a pair of values for a and b, where a and b are positive unit fractions, that make $\frac{1}{a} > \frac{1}{b}$ true.

Student activity: equations and inequalities

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O $a = 1$ and $b = 1$

O $a = -1$ and $b = 1$

O $a = 9$ and $b = 3$

Question 2

$100 = 26 + \frac{37}{n}$

What value of n makes this equation true?

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Question 3

$3x^2 < 81$

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Question 4

$-200 > \frac{5a}{4}$

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O $a = -160$

O $a > -160$

O $a < -160$

O $a = 160$